DB Week 11 Workshop

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Workshop Overview

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Data Warehouse

02

Lab: Database Admin

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Definition

02

Terminology

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Dimension
Modelling

Data Warehouse

Data Warehouse

- **Single** database of organisational data
  - Integrates data from multiple sources
  - Makes data available to managers / users
  - Supports analysis & decision-making

- UNLIKE ER models:
  - **ER**: Data organised around ***conceptual entities***
  - **DW**: Data organised around ***business processes***
    - e.g. sales, finance, or marketing

DW Characteristics

1. **Subject oriented**
   - Organised around particular subjects, e.g. sales, customers, products
2. **Validated, integrated data**
   - Data from different systems → *common format* → *validated before storing*
   - Allows comparison & consolidation of different data
3. **Time variant**
   - Historical data, timestamped → allow for trend analysis etc.
4. **Non-volatile**
   - Users have read access only
   - All updating done automatically

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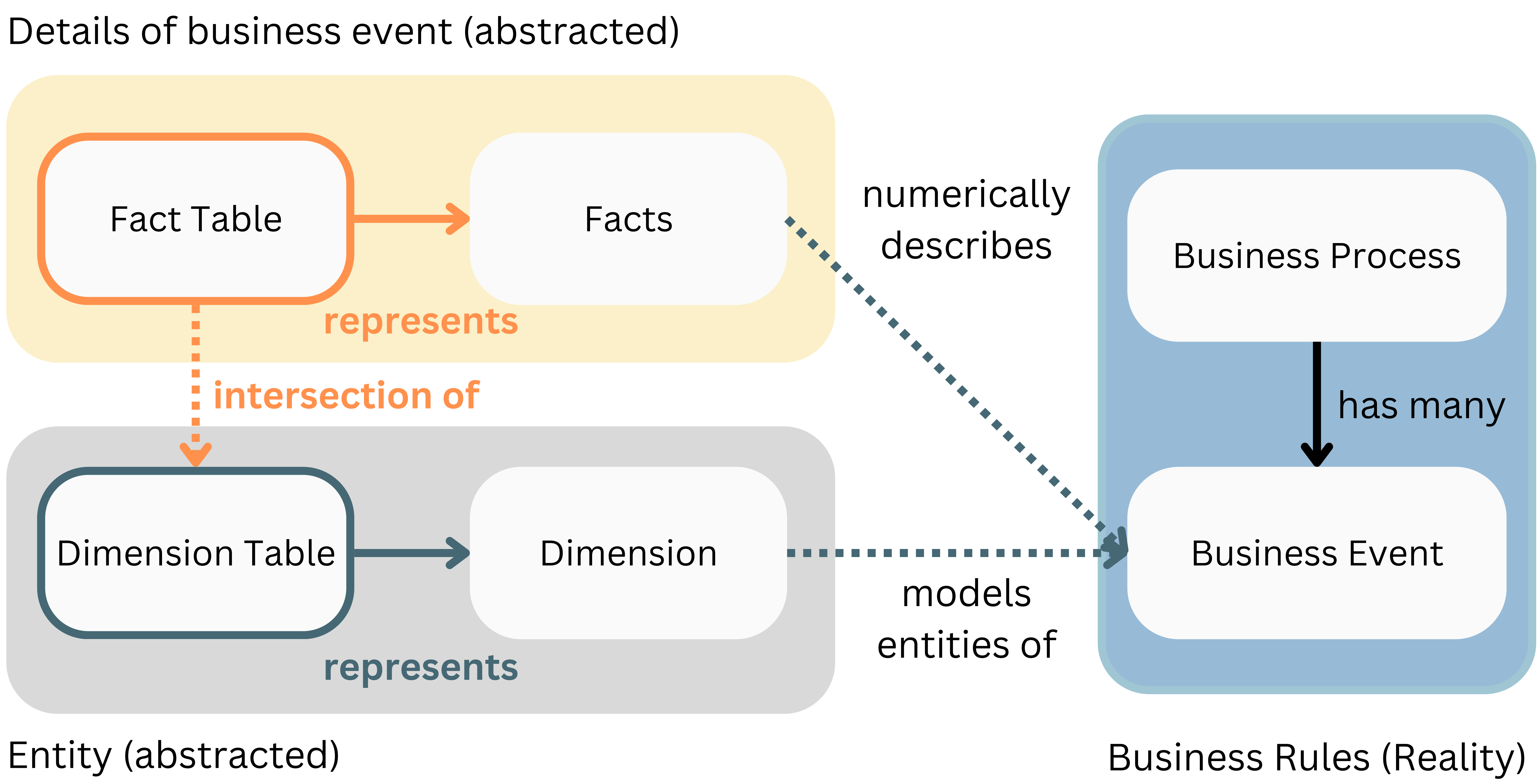
Dimension Modelling

Data Warehouse



Details of business event (abstracted)
Fact Table
Facts
represents
intersection of
numerically
describes
Business Process
has many
Business Event
Dimension Table
Dimension
represents
models
entities of
Entity (abstracted)
Business Rules (Reality)

Business Events

- An event that occurs as part of a *business process*

- Examples:
  - Sales business process → an individual order or sale
  - Finance → a payment
  - Marketing → a view of a webpage or a click on an online ad

Dimension

- An <u>entity</u> that ***describes*** and gives ***context*** to a business event
  - Entity that models business events, stores relevant data as attributes
  - Common dimensions: Time, customers, products, locations

- Example (business event):
  - "A CEO is interested in a comparison of revenue of a **new model** of the product with the **older model** in **every quarter** of the year by **customer demographic** group"
  - Dimensions: **Time**, **Product**, **Customer**

Dimension Table

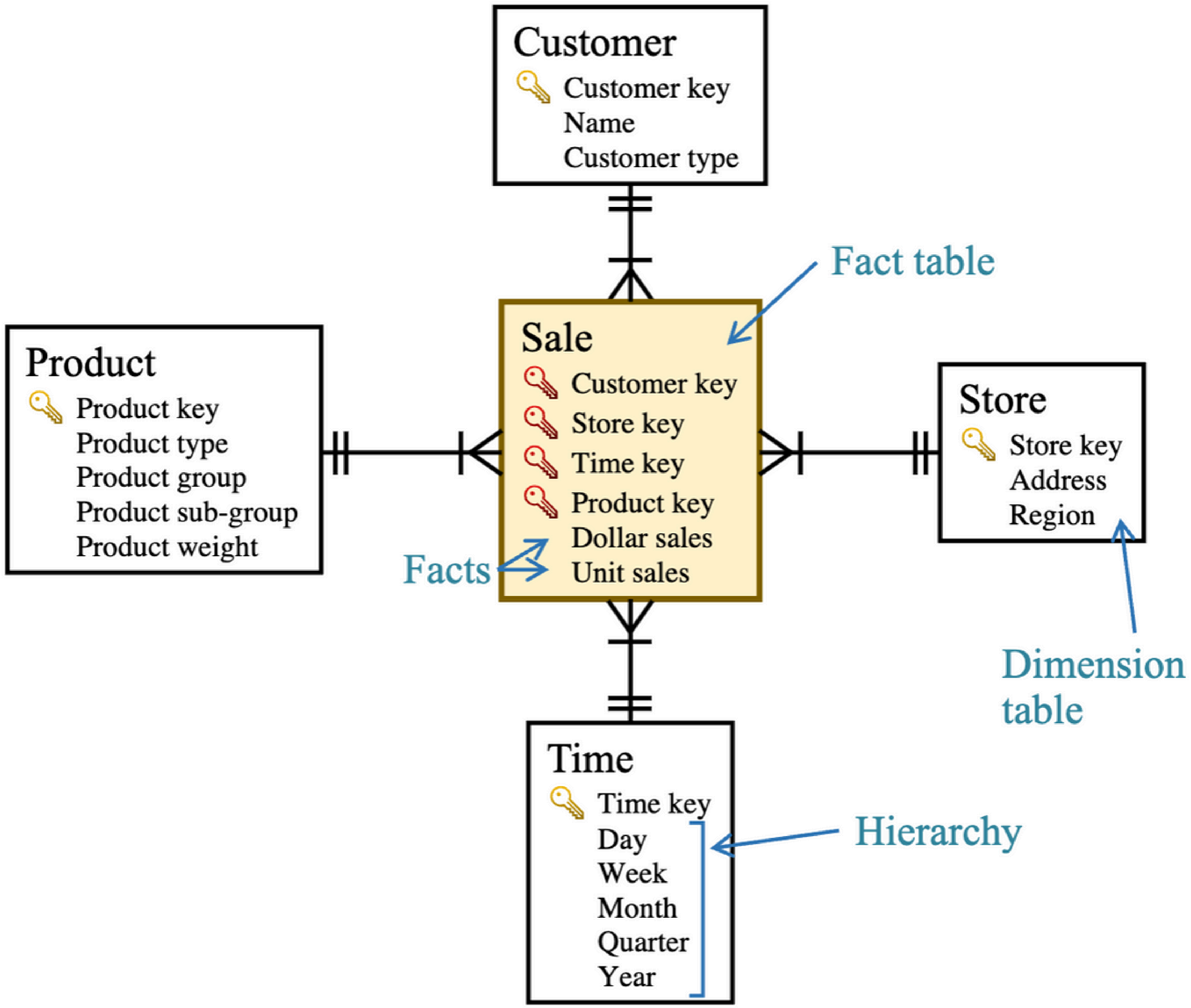
- Dimensions are represented as **dimension tables**, storing attributes

- ***Hierarchy***:
  - Hierarchy of attributes describing a dimension
  - Sequence of attributes across different levels of detail

- Examples:
  - Time → day, week, month, quarter, year ...
  - Location → city, state, country ...

Dimension Table

- Customer
- Store
- Time
- Product



Fact

- ***Numeric*** measurement of a meaningful and significant business event

- Example: *Customer* buys a *product* at a certain *location* at a certain *time*
    - facts: revenue, #items sold, total profit earned

- Stored in a ***fact table*** as attributes

Fact Table

- An intersection of the dimensions
  - PK = FKs referencing the dimension tables
  - Other attributes = facts

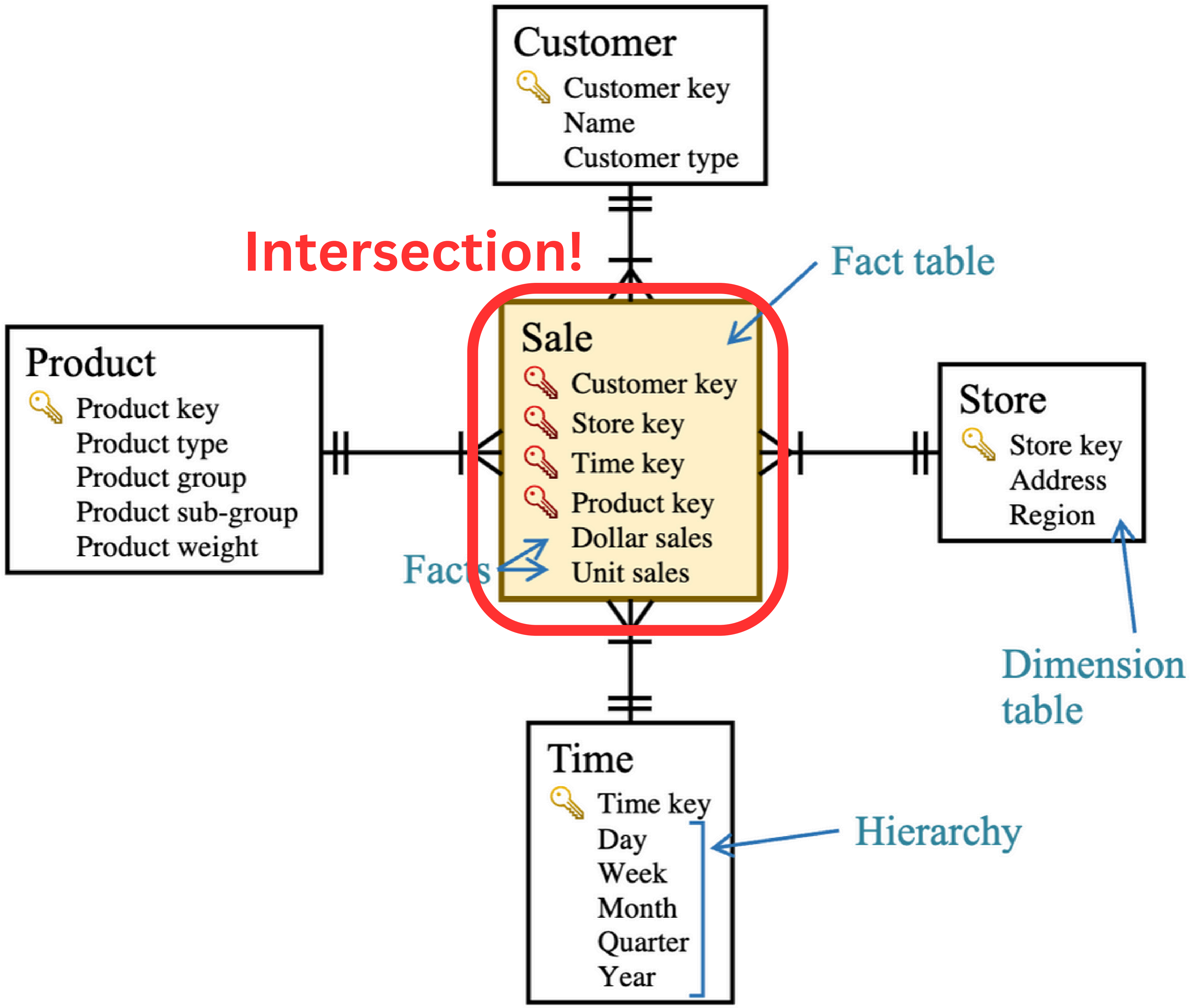
- ***Granularity***: The level of detail present in a fact table
  - The finer the granularity, the more precisely a query can extract details

- Fact table can store each business event in own row, or many events aggregated together

Fact Table

- PK:
  - FK from dimension tables
- Facts:
  - dollar sales
  - unit sales

- What are the granularities?
  - Finest level of detail



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Dimension Modelling

Data Warehouse

Modelling

- Star Schema
  - Simple and restricted type of ER model

- Dimensional model consists of:
  a. Fact table
  b. Several dimension tables
  c. (Optional) hierarchies in dimensions

Process

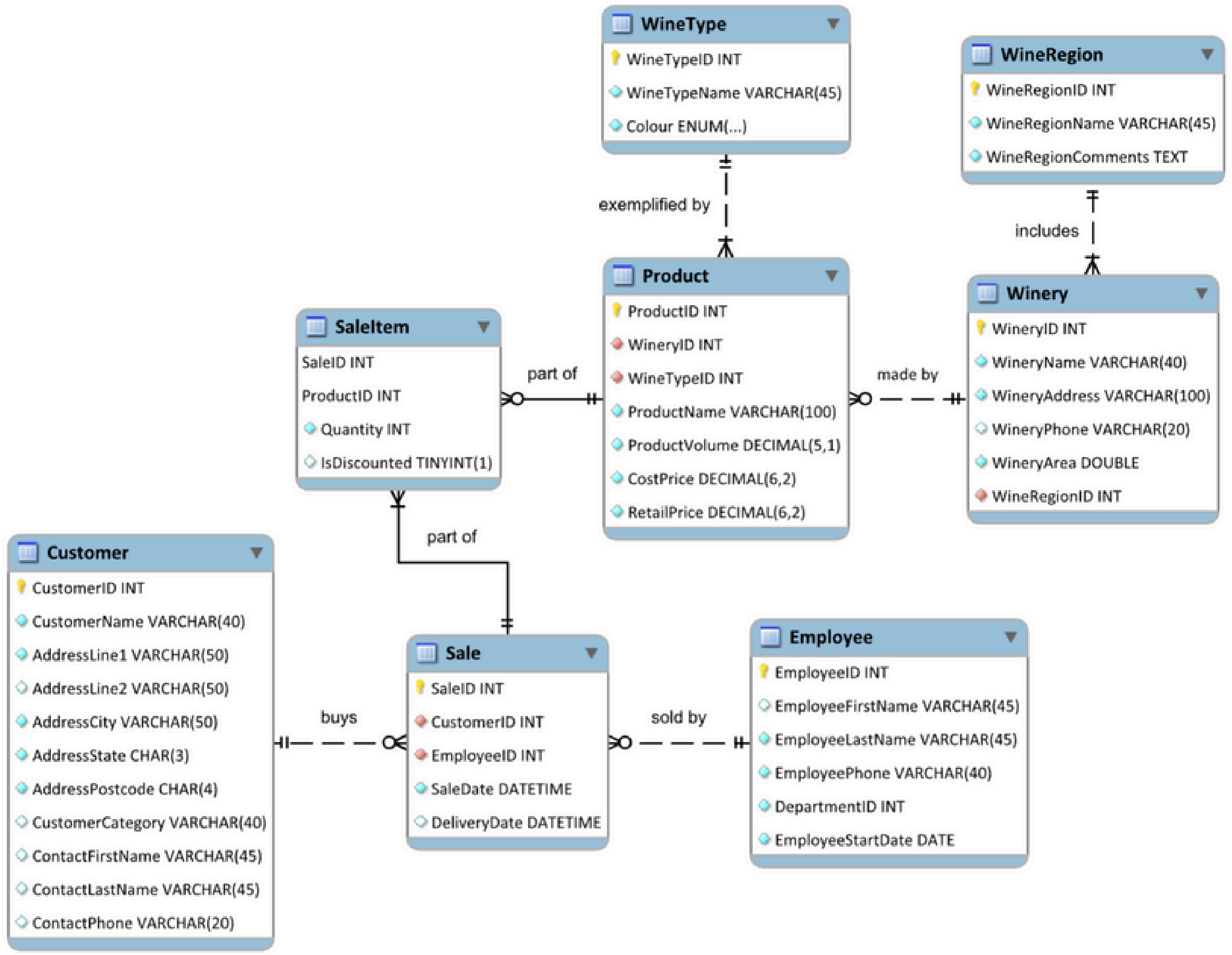
1. Choose a **business process**
2. Choose the measured **facts**
    - Usually numeric, additive quantities
3. Choose the **granularity** of the fact table
4. Choose the **dimensions**
5. Complete the **dimension tables**

Q1.

Wimmera Wines is a large company that takes deliveries of grapes from wine growers, produces and bottles wine, and sells those bottles to retailers and restaurants. They produce many different types of wine at a range of price points, from cheap cask wine to top-of-the-range vintage bottles.

The company is aiming to increase their product sales by 20% in comparison to the last 3 years. To help the business achieve their aim, you have been hired to design a data warehouse that can help business managers analyse data related to the sales theme.

The company is keen to understand all the aspects of their business that contribute to strong sales. For example, two business measures that have been mentioned are “total number of units of each product sold” and “revenue generated by each employee per year”.



Q1.(a) As a class, brainstorm some more business measures that Wimmera Wines managers might need if they are to achieve their aim.

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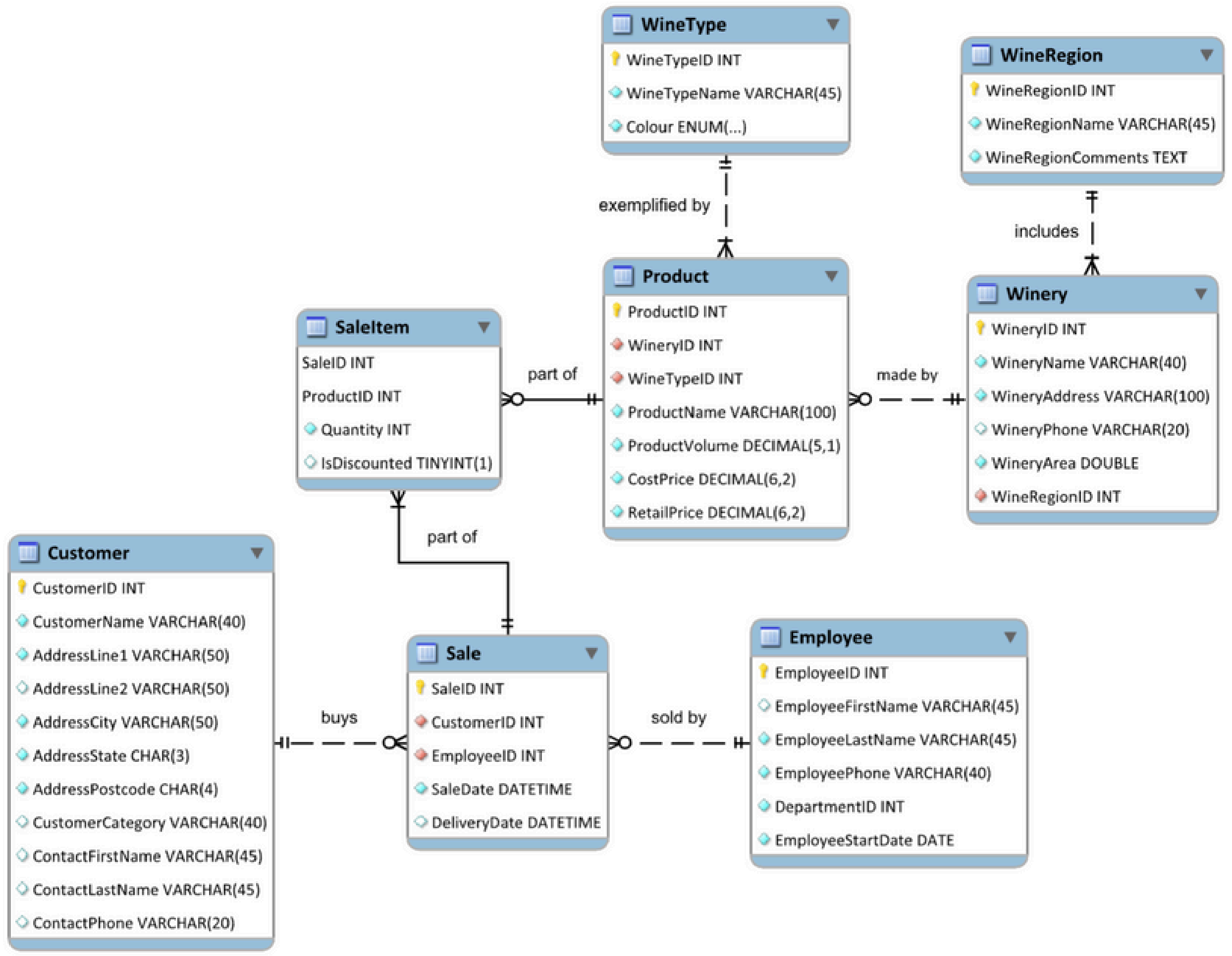

Q1.(a) As a class, brainstorm some more business measures that Wimmera Wines managers might need if they are to achieve their aim.

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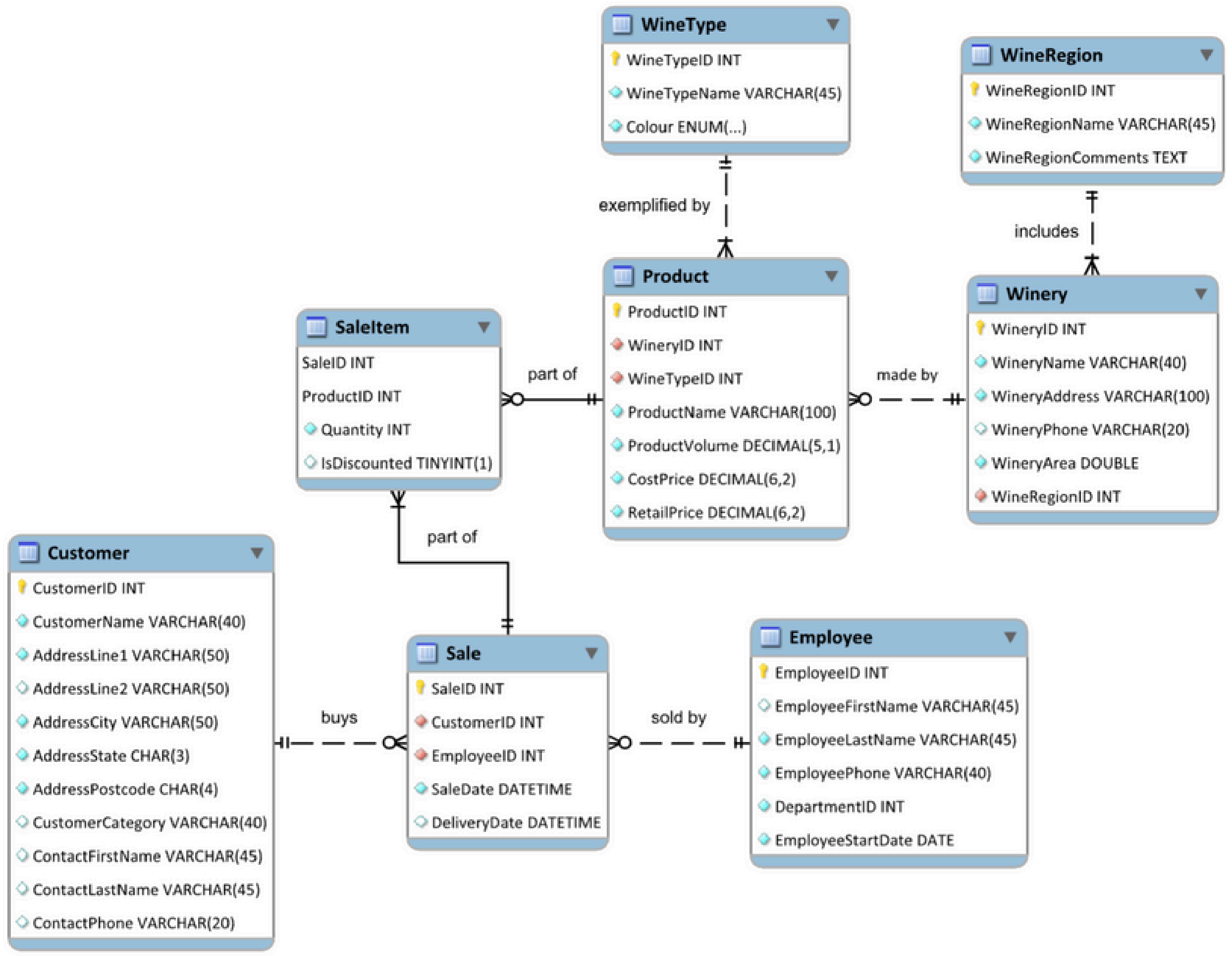

- Number of products sold per year
- Sales by a particular state
- Sales of a product in a given quarter of a year
- Revenue generated from a particular customer category
- Which product is selling the best (hence generating the most revenue)?

Q1.(b)

The company is aiming to increase their product sales by 20% in comparison to the last 3 years. To help the business achieve their aim, you have been hired to design a data warehouse that can help business managers analyse data related to the sales theme.

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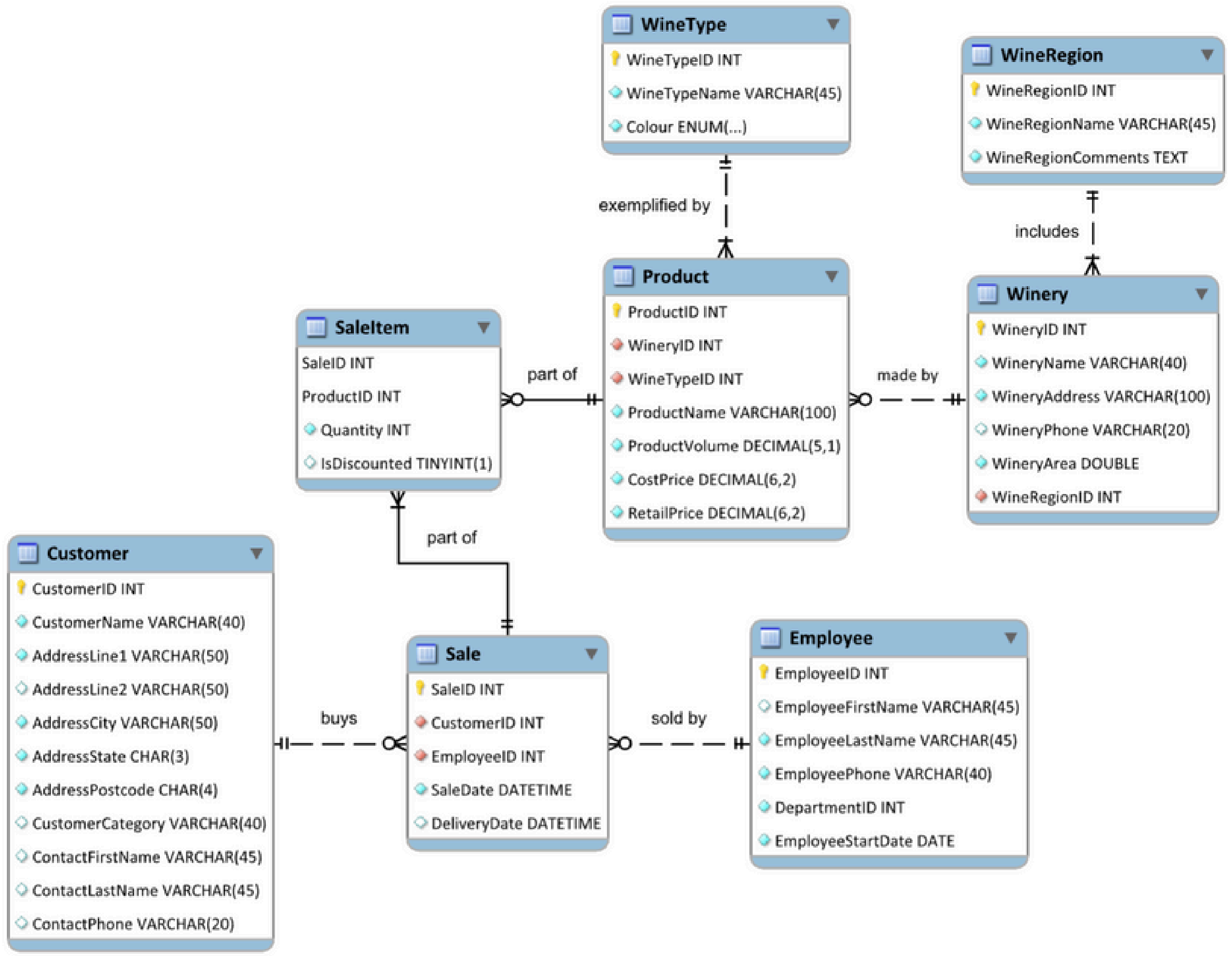


Use Kimball’s five-step dimensional design process to design a dimensional model for Wimmera Wines’ product sales subject area.

1. Select and explain the business process.
2. Identify and explain the facts.
3. Declare the grain and justify your choice.
4. Identify and explain the dimensions.
5. Complete the dimension tables

Q1.(b) 1. Select and explain the business process.

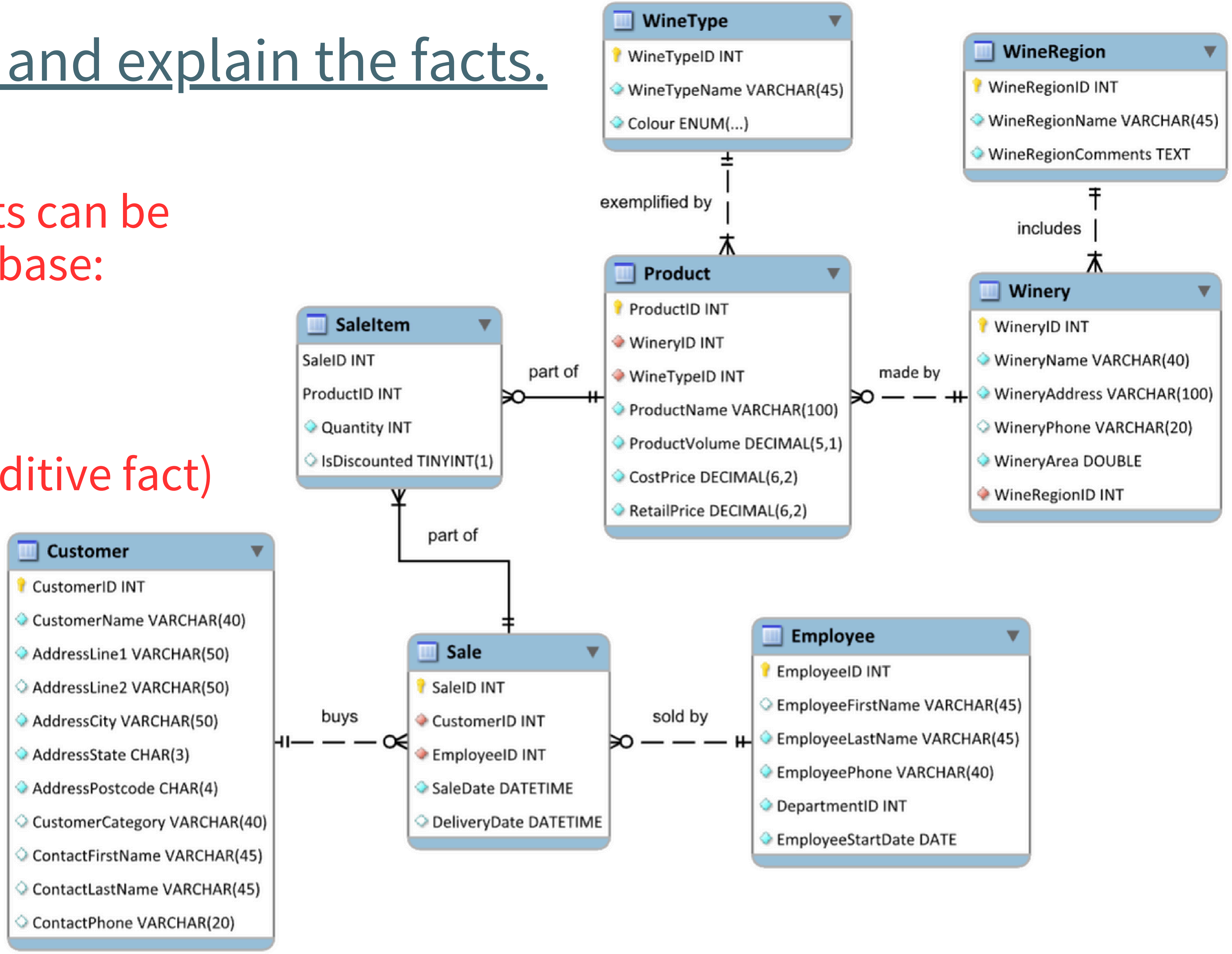
The company is keen to understand all the aspects of their business that contribute to strong sales. For example, two business measures that have been mentioned are “total number of units of each product sold” and “revenue generated by each employee per year”.

- Business process → **product sales**

Q1.(b) 2. Identify and explain the facts.

The following sales-related facts can be extracted from the source database:

- Unit sales
- Dollar sales
- Profit amount
- Discount indicator (non-additive fact)



Q1.(b) 3. Declare the grain and justify your choice.

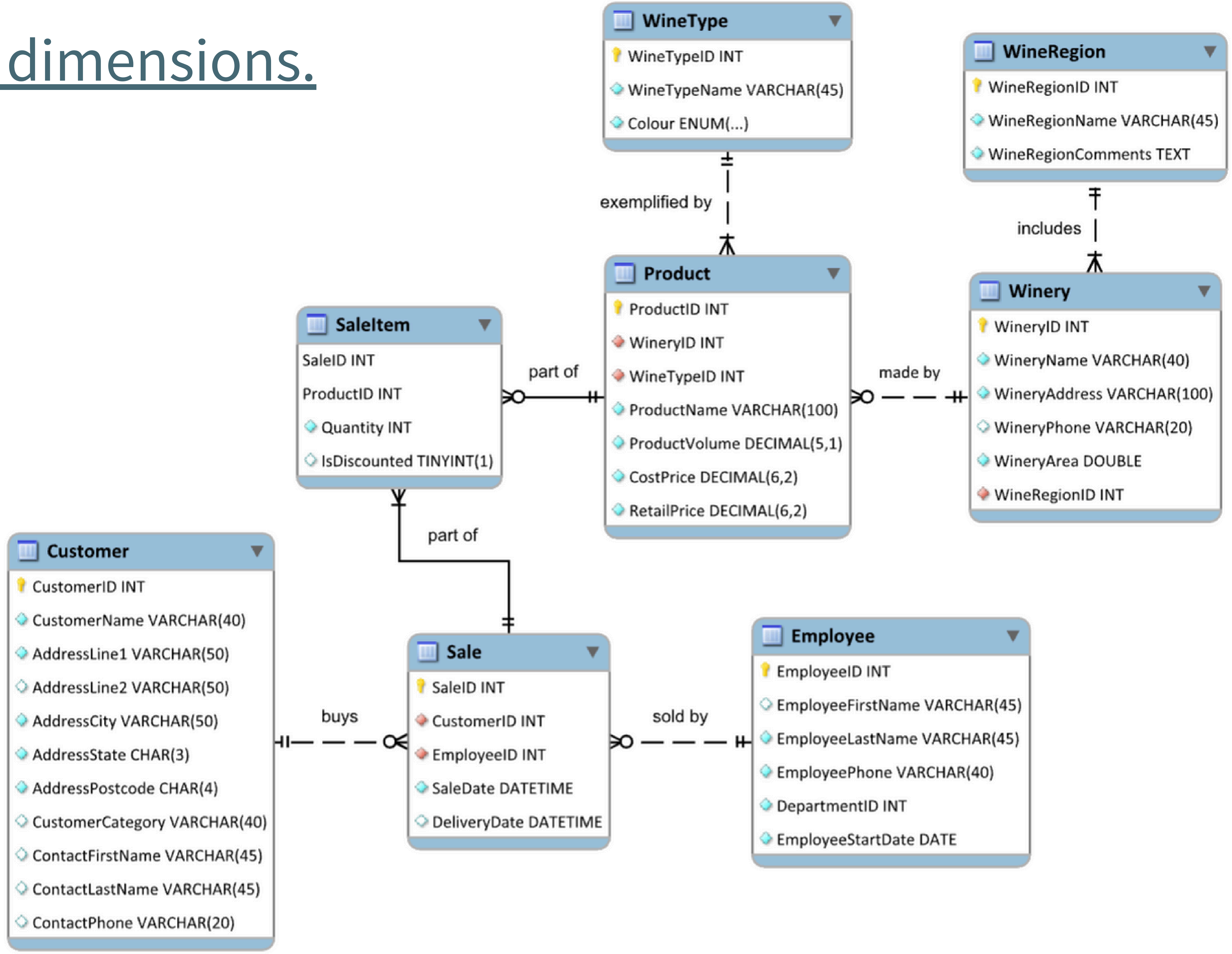
Wimmera Wines is a large company that takes deliveries of grapes from wine growers, produces and bottles wine, and sells those bottles to retailers and restaurants. They produce many different types of wine at a range of price points, from cheap cask wine to top-of-the-range vintage bottles.

- Would not make a large number of sales
- Each individual sale can include large quantities of items

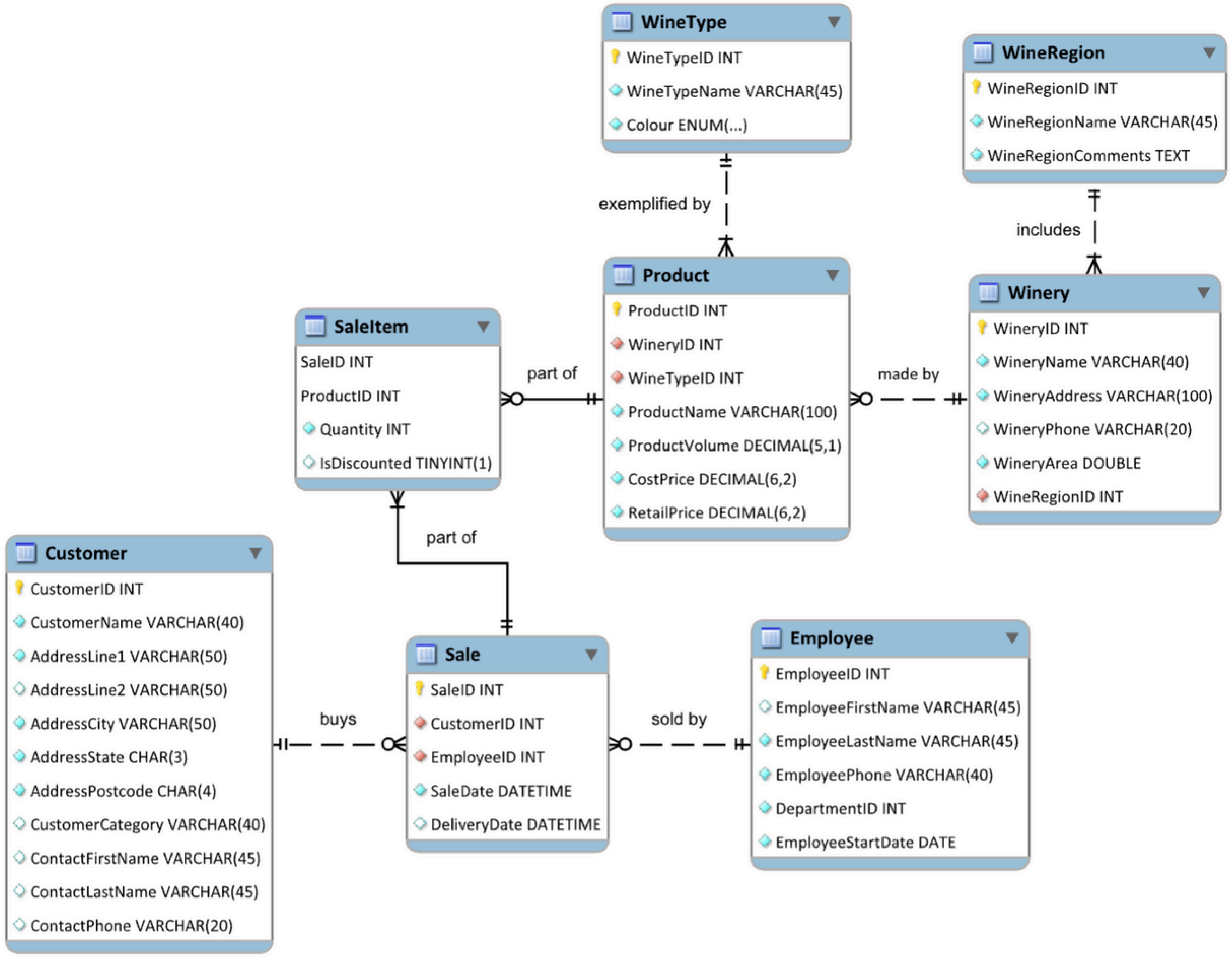
- Appropriate to store each sale item as its own row
- If do not require such detailed info, and perhaps weekly sales sufficient
  - → data would be aggregated by week

Q1.(b) 4. Identify dimensions.

- Employee
- Customer
- Time
- Product

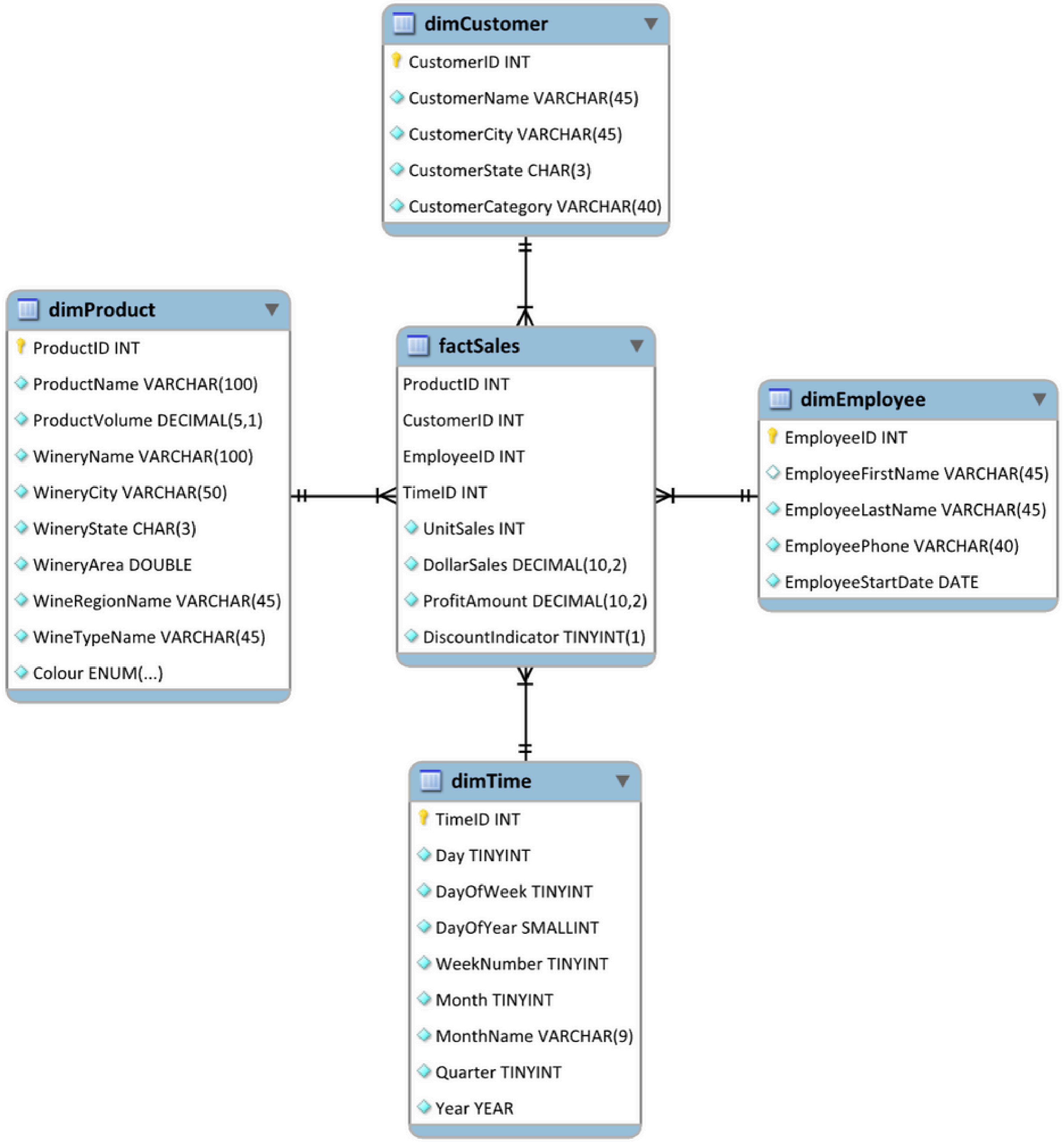


Q1.(b) 5. Complete the dimension tables



Q1.(b)

Notice how the Product dimension is denormalised. It has many transitive functional dependencies, such as WineryName → WineryCity and WineTypeName → Colour.



Q2.

Consider the following fact table:

Sale
- Time key
- Geography key
- Product key
- Dollar sales
- Unit sales

Suppose the following sales data has been extracted from the business's operational database:

SaleID	SaleDate	CustomerID	CustomerCity	ProductID	Price	Quantity
54	2003-12-13 14:13	788	Melbourne	9644	$10.00	2
54	2003-12-13 14:13	788	Melbourne	8574	$15.00	1
67	2003-12-13 15:05	903	Melbourne	9644	$10.00	1
76	2003-12-13 17:26	322	Sydney	9644	$5.00	4
77	2003-12-14 09:58	292	Melbourne	8229	$15.00	2

a. Starting from this source data, how many rows will be inserted into the fact table if an hourly grain is selected?

b. How many rows will be inserted into the fact table if a daily grain is selected?

c. At which level of granularity can we answer questions about hourly sales? At which level of granularity can we answer questions about daily sales?

Q2.(a)

Consider the following fact table:

Sale
- Time key
- Geography key
- Product key
- Dollar sales
- Unit sales

- None of these sale-item rows share the same hour, geography and product.
    - No aggregation can be performed.
- Five rows will be inserted into the fact table

Suppose the following sales data has been extracted from the business's operational database:

SaleID	SaleDate	CustomerID	CustomerCity	ProductID	Price	Quantity
54	2003-12-13 14:13	788	Melbourne	9644	$10.00	2
54	2003-12-13 14:13	788	Melbourne	8574	$15.00	1
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a. Starting from this source data, how many rows will be inserted into the fact table if an hourly grain is selected?

Q2.(b)

Consider the following fact table:

Sale
Time key
Geography key
Product key
Dollar sales
Unit sales

- Rows w/ same keys will be aggregated
  - 54 & 67
- → 4 rows will be inserted in total

Suppose the following sales data has been extracted from the business's operational database:

SaleID	SaleDate	CustomerID	CustomerCity	ProductID	Price	Quantity
54	2003-12-13 14:13	788	Melbourne	9644	$10.00	2
54	2003-12-13 14:13	788	Melbourne	8574	$15.00	1
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77	2003-12-14 09:58	292	Melbourne	8229	$15.00	2

b. How many rows will be inserted into the fact table if a daily grain is selected?

Q2.(c)

Consider the following fact table:

Sale
- Time key
- Geography key
- Product key
- Dollar sales
- Unit sales

- Hourly sale: a hourly grain or finer
- Daily sale: a daily grain or finer

Suppose the following sales data has been extracted from the business's operational database:

SaleID	SaleDate	CustomerID	CustomerCity	ProductID	Price	Quantity
54	2003-12-13 14:13	788	Melbourne	9644	$10.00	2
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c. At which level of granularity can we answer questions about hourly sales? At which level of granularity can we answer questions about daily sales?